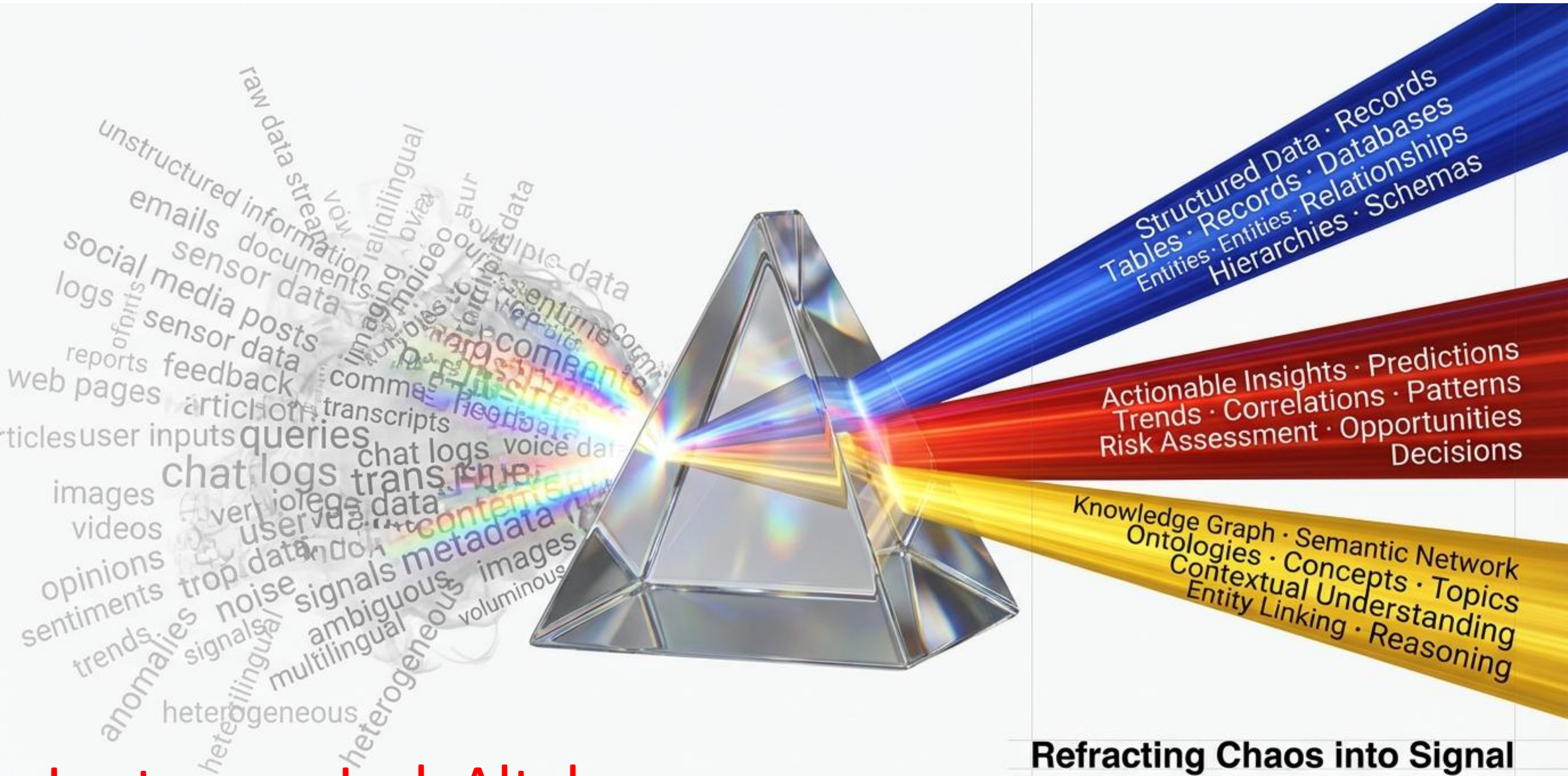
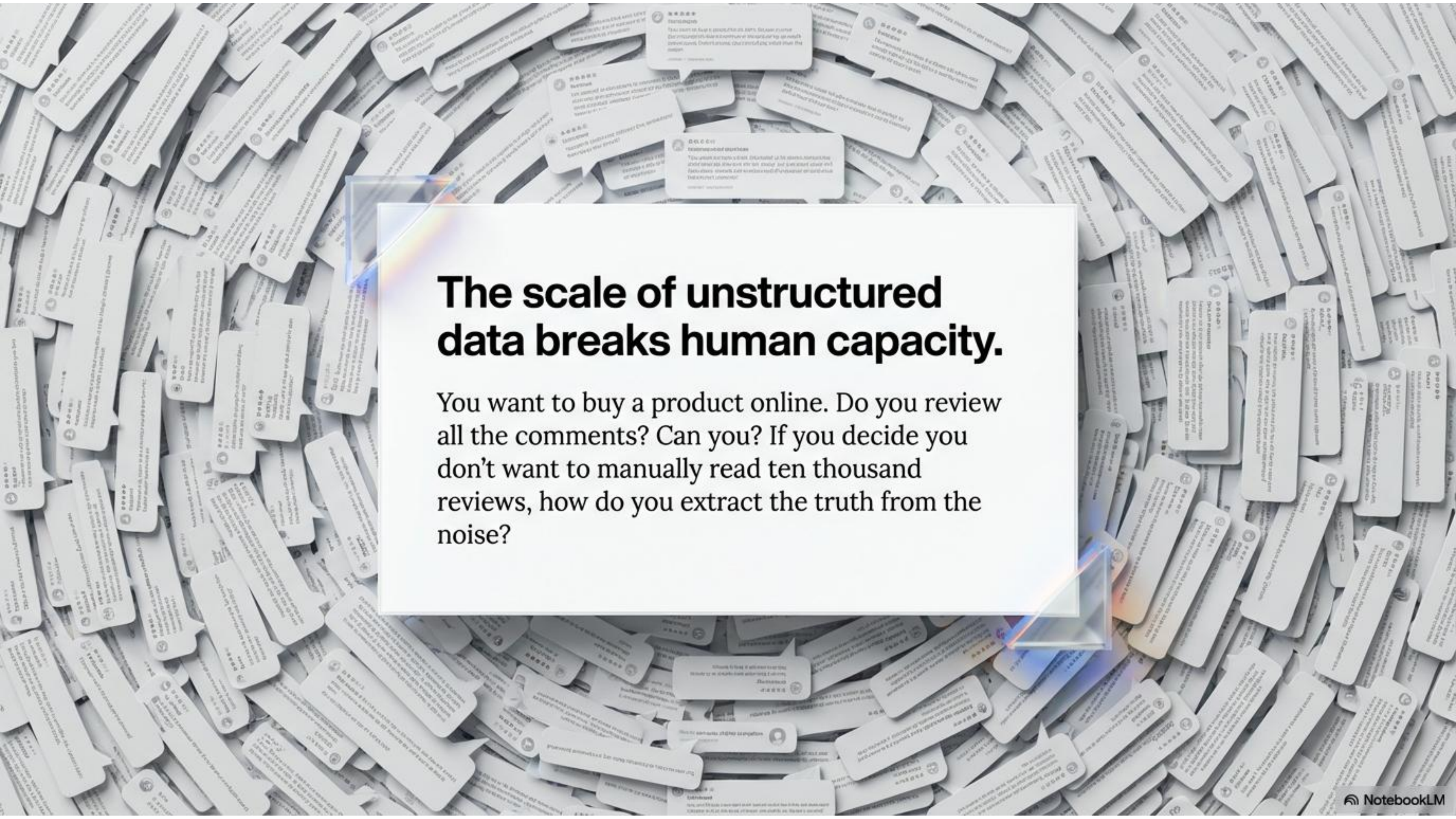




Lecturer – Jack Altal

Refracting Chaos into Signal

A pragmatic guide to Data Mining, Text Mining, and Natural Language Processing.



The scale of unstructured data breaks human capacity.

You want to buy a product online. Do you review all the comments? Can you? If you decide you don't want to manually read ten thousand reviews, how do you extract the truth from the noise?

Defining the boundaries of the analytical lexicon.



Data Mining (DM)

Input: Structured Data (Tables, Rows, Integers).

Outcome: Transforms structured data into actionable insights.



Text Mining (TM)

Definition: The process of distilling actionable insights from text.

Input: Unstructured Data (Reviews, Emails, Notes).

Outcome: Transforms unstructured text into actionable insights.



Natural Language Processing (NLP)

Input: Large-scale Linguistic Data.

Outcome: Utilizes big models capable of reasoning like humans to in models capable of reasoning like humans to parse complex grammatical meaning.

Unstructured text holds the blueprint to organizational behavior.



 Download HR: Understand candidate perceptions and algorithmically match job descriptions with resumes.	 Download Marketing: Measure campaign effectiveness, identify brand evangelists, and build customer propensity models.	 Download Health: Analyze doctor's notes to understand patient readmission characteristics and utilize medical charts for predictive modeling.
 Download Law: Automatically classify vast repositories of legal documents and case files as relevant or irrelevant.	 Download Customer Service: Identify actionable social media posts, classify support tickets as urgent, and generate sentiment-based recommendations.	 Download Fraud Detection: Monitor unstructured communications to detect anomalous patterns indicative of financial crimes.

Unstructured text holds the blueprint to organizational behavior.





altalsec

Edit profile

Popular repositories

OpenCV

Public

OpenCV build for NVIDIA GeForce RTX 5080

● CMake

Anomaly-Detection

Public

Cyber security based datasets for DT/DM/ADT testing

● HTML

12 contributions in the last year

Contribution settings

2026

	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	
Mon													■
Wed													
Fri											■		

Learn how we count contributions

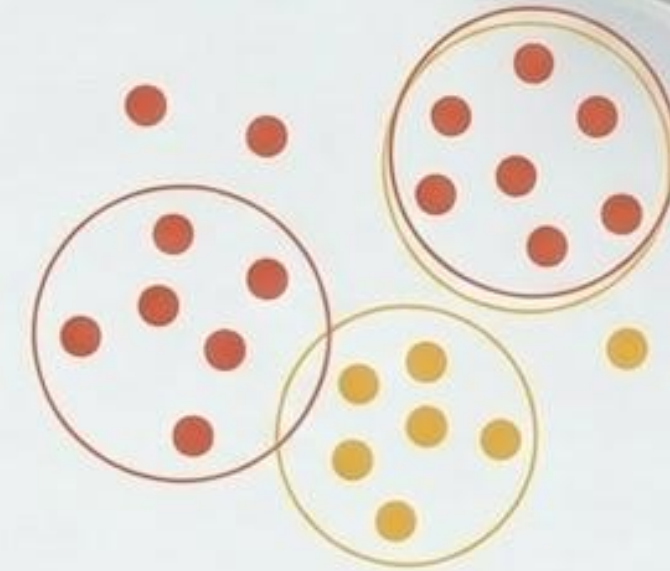
Less ■ ■ ■ ■ More

Two distinct paths to algorithmic insight.



Supervised Learning

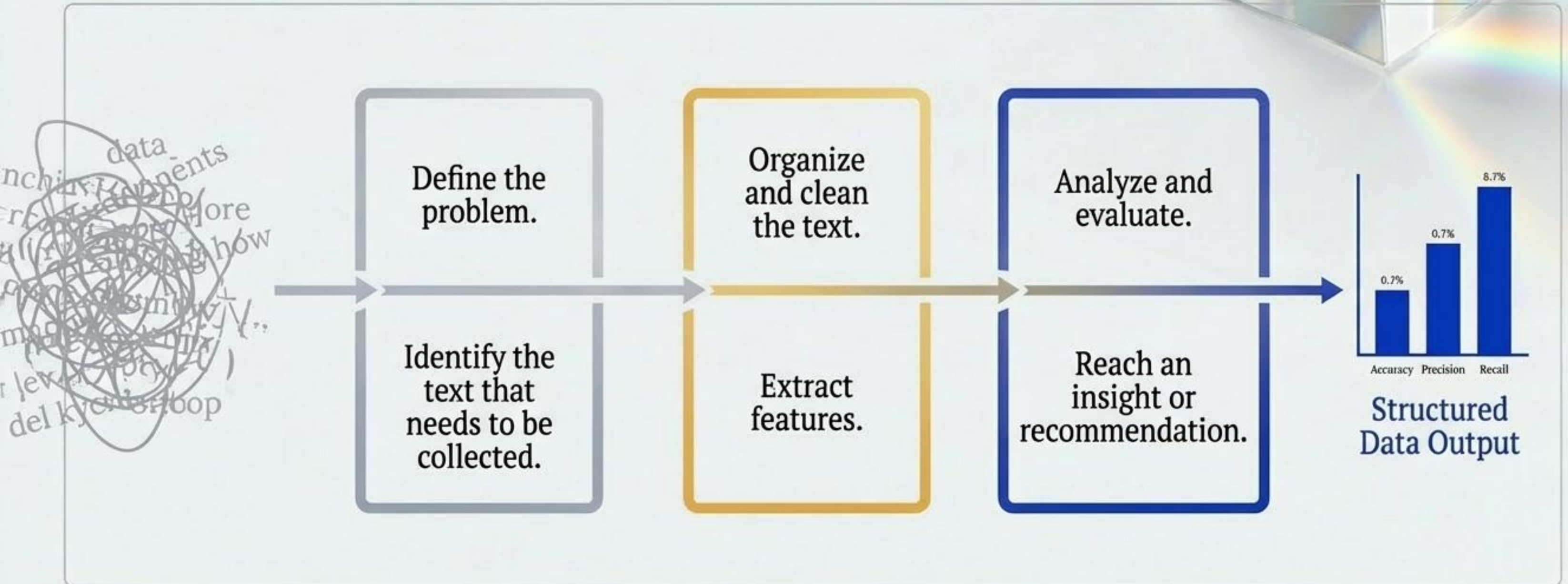
- **The Condition:** The outcome (Y) is known.
- **The Goal:** Prediction.
- **The Techniques:** Regression, Classification, Neural Networks.



Unsupervised Learning

- **The Condition:** The outcome (Y) is unknown.
- **The Goal:** To find hidden patterns or groups.
- **The Techniques:** Cluster Analysis, Topic Modeling.

The Text Mining Workflow.



Ingesting raw data and structuring the vocabulary.

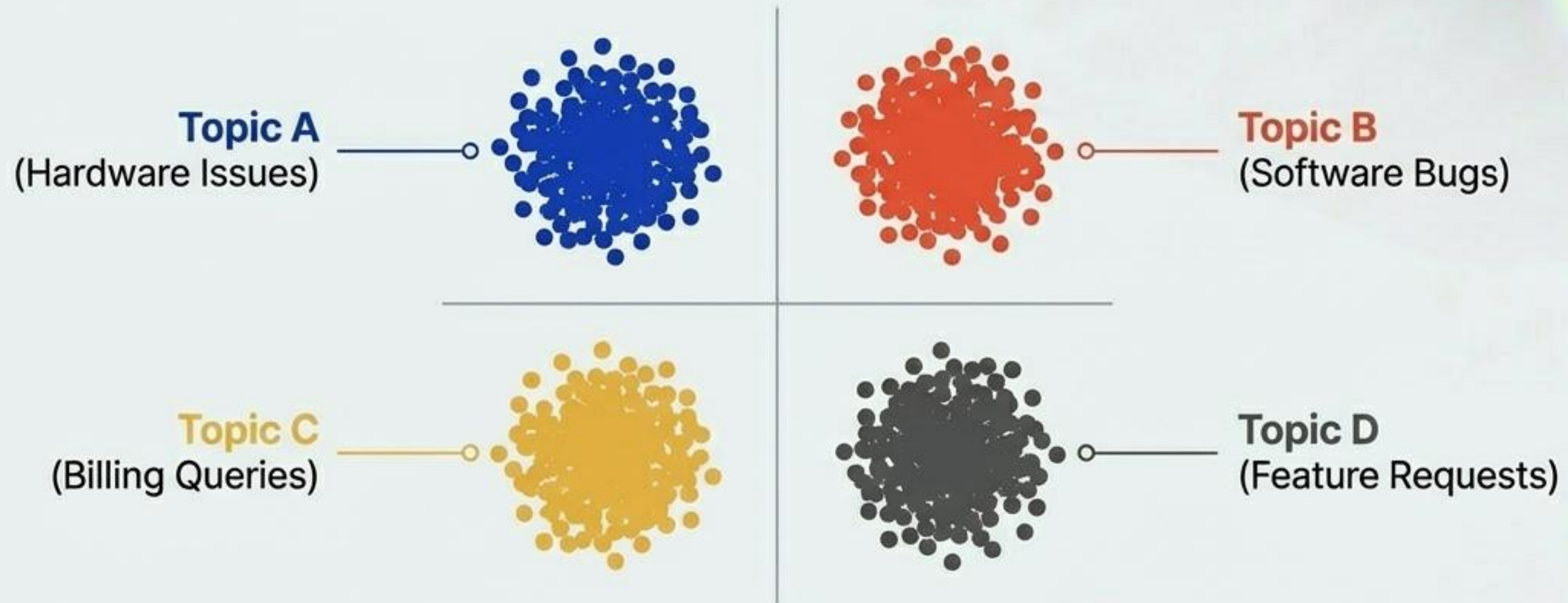
Algorithms cannot read sentences. We extract text using APIs and transform it into a mathematical representation through tokenization.



Word Token	Doc 1 Count	Doc 2 Count
breach	14	0
firewall	7	2

Unsupervised Analysis identifies hidden groupings.

When the outcome is unknown, we deploy clustering and topic modeling. The algorithm scans the text representation tables and groups documents based on mathematical proximity.

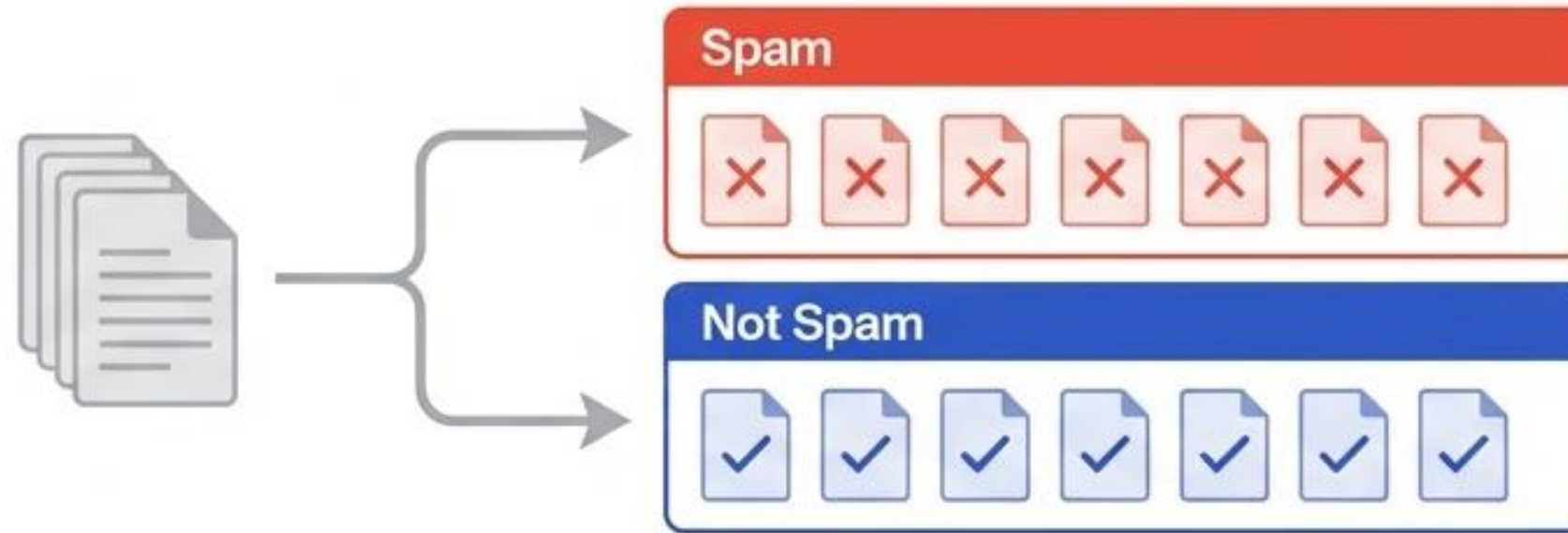


Supervised Analysis predicts specific outcomes.

Text analysis for predictive modeling operates when historical outcomes are known, allowing us to build models that classify new data or predict continuous variables.



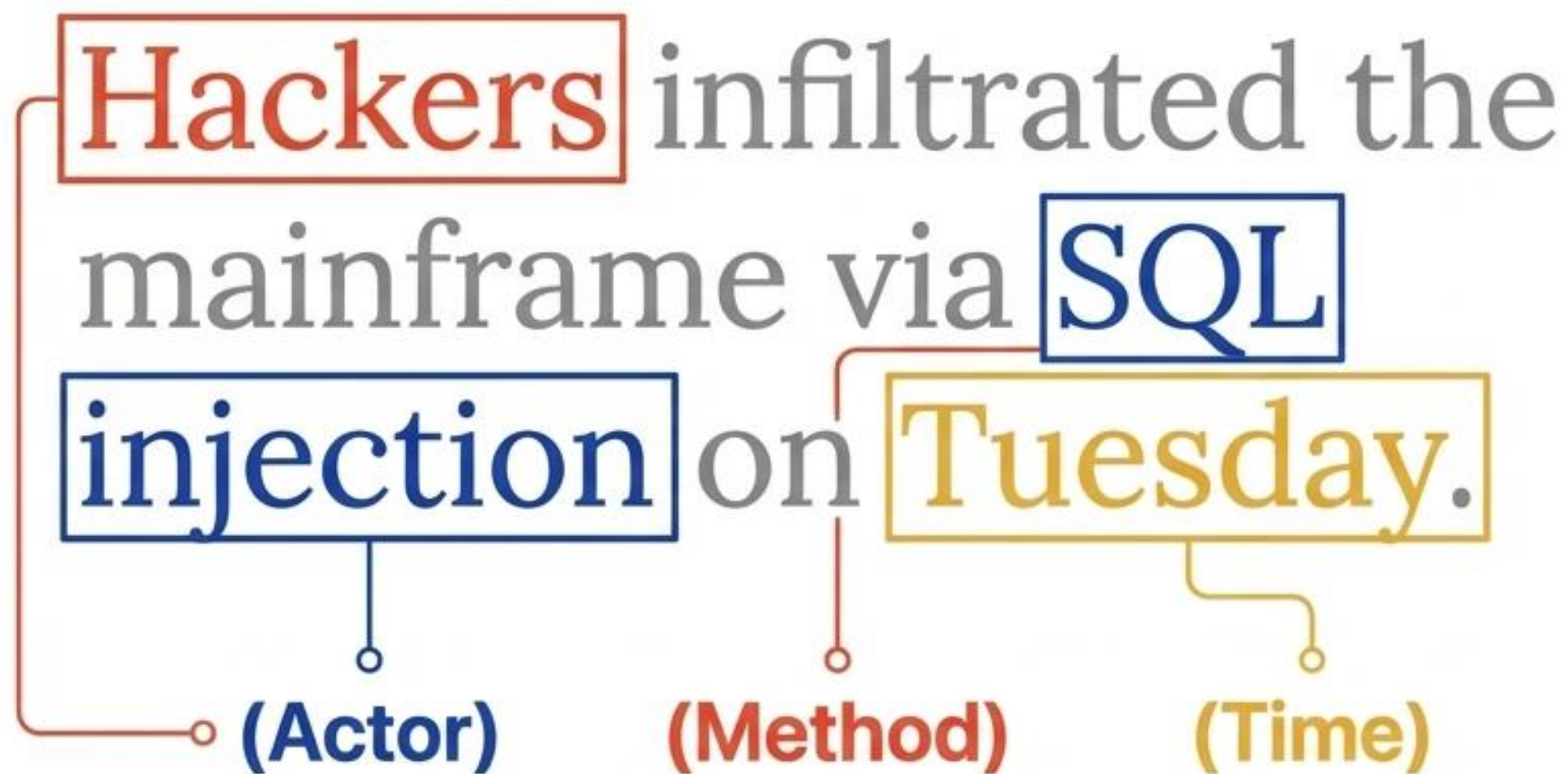
Classification (Categorical outcomes)



Regression (Continuous prediction)



Natural Language Processing and Named Entity Recognition.



Core NLP Tasks

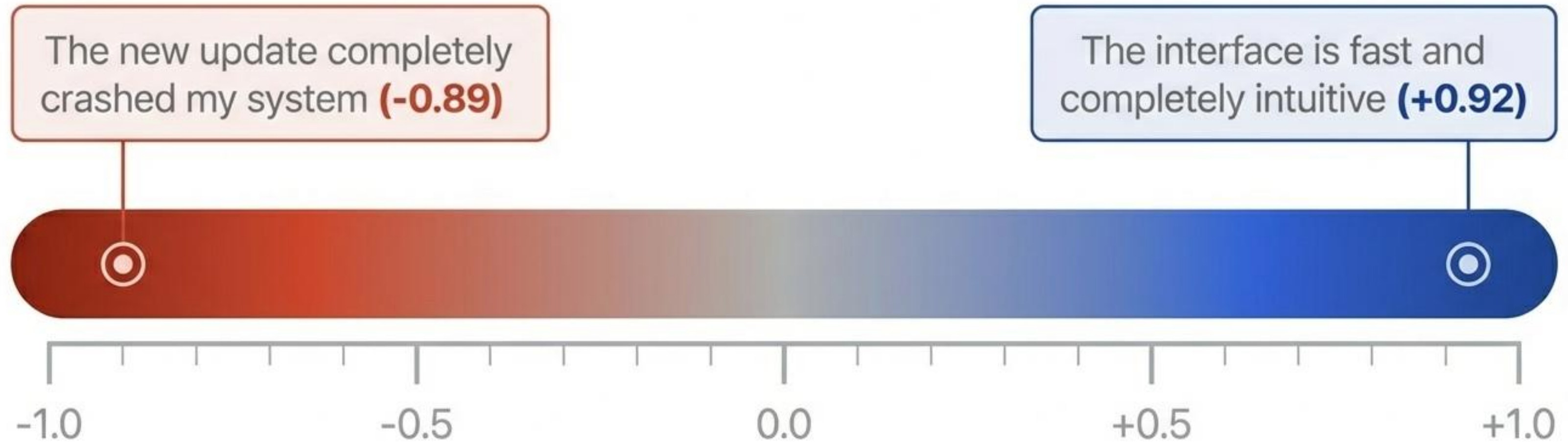
Mapping grammar structure to extract entities utilizing POS tags:

PRP (Pronoun),
VBZ (Verb),
NNS (Noun plural),
IN (Preposition),
DT (Determiner),
NN (Noun singular).

Quantifying emotion through Sentiment Scoring.

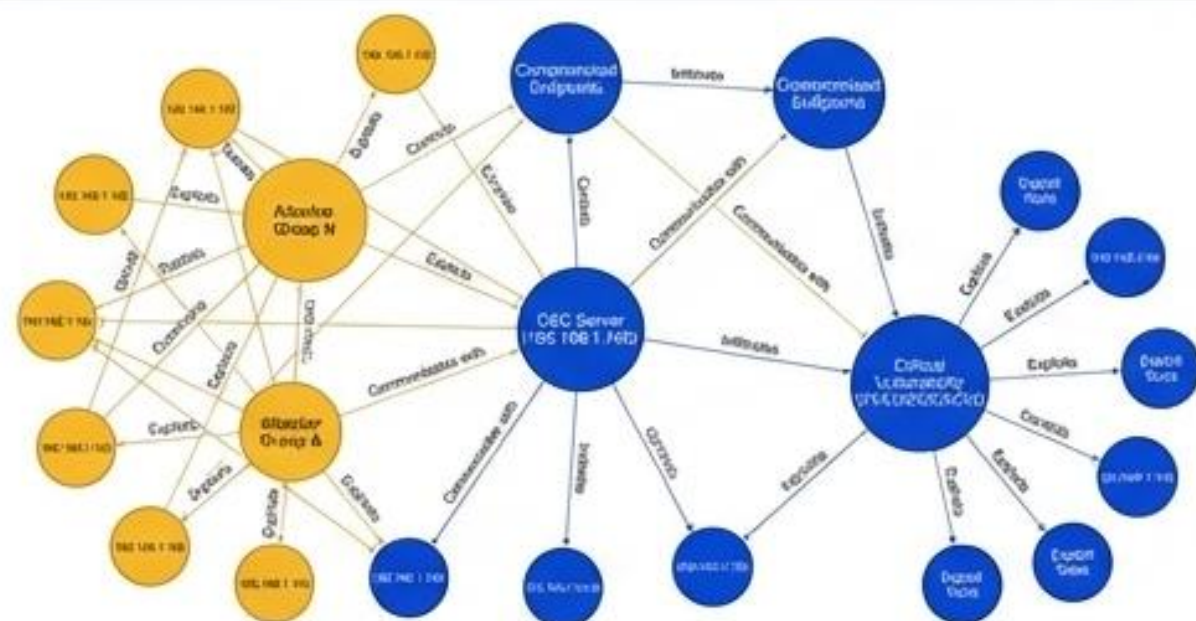


Sentiment analysis algorithms assign a mathematical polarity score to unstructured text by evaluating the context, modifiers, and specific lexicon used within a document.

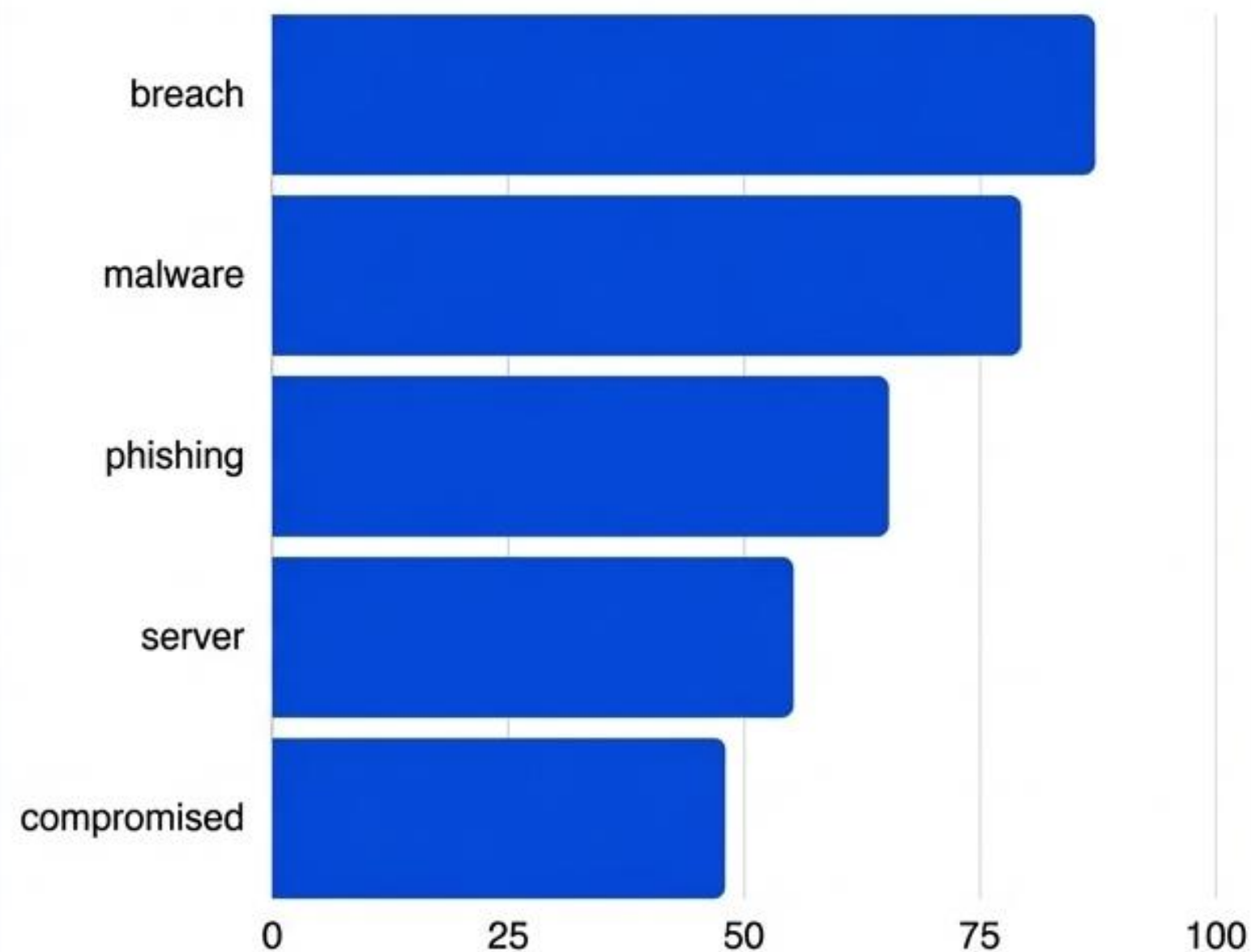


Visualizing the final signal

Can I understand the main issues with this incident at a single glance?

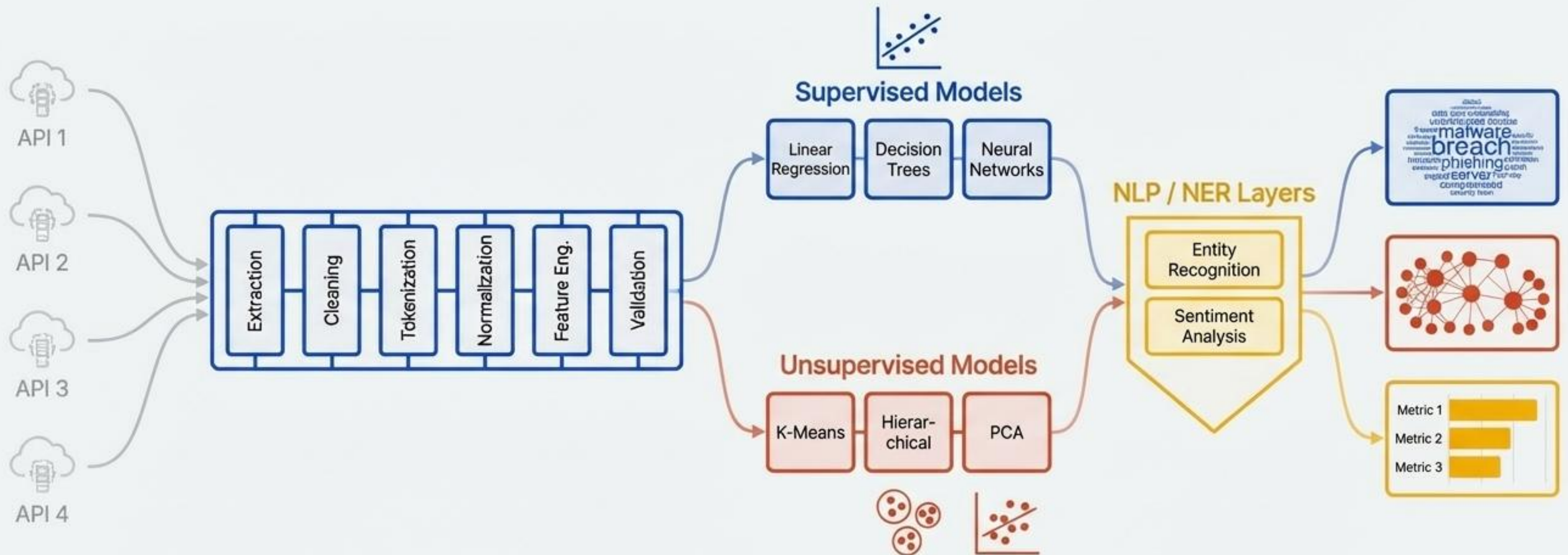


Top 5 Key Terms Frequency in Incident Report



The Unstructured Data Ecosystem.

From raw noise to structured signal. Every API extraction, tokenization table, learning model, and visualization is a single gear in a larger mechanism designed to solve complex business problems.



Mission 01: Discover the Unknown.



[k]

Objective: Practice Unsupervised Learning on Kaggle.

Action: Navigate to Kaggle and select a raw, unstructured text dataset (e.g., Reddit comments or Amazon reviews). Deploy a clustering or topic modeling algorithm to discover three distinct, hidden groupings within the data. Visualize your findings using a word cloud or bar graph.

Mission 02: Predict the Future.



Objective: Practice Supervised Learning and Sentiment Scoring.

Action: Find a [Kaggle dataset](#) containing text with known outcomes (e.g., [labeled spam emails](#) or [movie reviews with 1-5 star ratings](#)). Clean the text, extract the features, and build a [classification model](#) that successfully [predicts the outcome](#) of new, unseen text entries based on their sentiment.

03

Mission 03: Extract the Entities



Objective: Practice **Natural Language Processing**.

Action: Select a complex dataset on Kaggle (e.g., financial news headlines or medical abstracts). Utilize an NLP library to execute **Named Entity Recognition (NER)**. Identify and extract specific nouns, actors, locations, or methods, proving your ability to pull highly structured, human-like reasoning from a wall of raw text. The noise is yours to control.
